Handling missing data

MCAR (Missing Completely At Random) means data is missing with no relation to any variable — purely random.
Example: a survey form lost due to printing error.
It's the best-case scenario for missing data because it doesn't bias results.

MAR (Missing At Random) means data is missing for a reason related to observed variables, not the missing one itself.
Example: Older people skip an income question more often — missingness depends on age, which we know.
It's manageable using statistical methods like imputation or regression.

MNAR (Missing Not At Random) means data is missing because of the value that's missing itself.
Example: People with very high income don't report their income — missingness depends on the hidden value.
It's the hardest type to handle since you can't fix it without making assumptions or collecting more data.

SMOTE :
SMOTE creates synthetic (new) examples of the minority class rather than simply duplicating existing ones.

⚖️ Why SMOTE is Needed

In classification problems, an imbalanced dataset looks like this:

Class	Count
0(majority)	900
1(minority)	100

→ If you train a model on such data, it might just predict all zeros — because that gives it 90% accuracy!
SMOTE helps fix this by creating more samples for class 1, so the model learns both classes properly.

🔬 How SMOTE Works (Step-by-Step)

1. Pick a sample from the minority class.
2. Find its k nearest neighbors (usually k=5) in the feature space.
3. Randomly select one of these neighbors.
4. Generate a new synthetic sample between the two points
   $x_{new}=x_i+rand(0,1)\times(x_{neighbor}-x_i)$
5. Repeat until the desired balance is reached.

So SMOTE interpolates between real samples to create synthetic but realistic new samples.

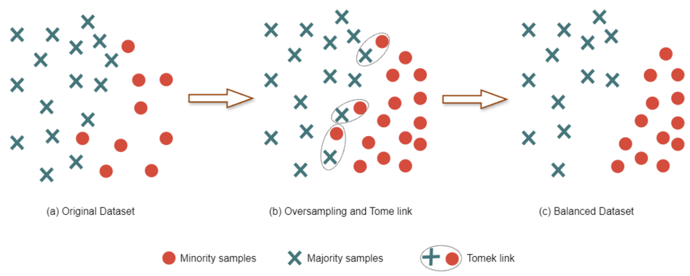
→ Imbalanced datasets impact the performance of the machine learning models and the Synthetic Minority Over-sampling Technique (SMOTE) addresses the class imbalance problem by generating synthetic samples for the minority class.

Data Imbalance in Classification Problem

→ Data imbalance in classification refers to skewed class distribution, hindering machine learning models' performance.

→ Techniques like Oversampling, Undersampling, Threshold moving, and SMOTE help address this issue.

→ Handling imbalanced datasets is crucial to prevent biased model outputs, especially in multi-classification problems.



HOW SMOTE WORKS